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Shear Modulus and Damping Ratio of Cohesive Soils

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This article presents results from a laboratory investigation into the dynamic characteristics of reconstituted and undisturbed cohesive soils by means of resonant-column tests. In particular, results showing the influence of various soil parameters, such as confining stress, overconsolidation ratio, void ratio, plasticity index, calcium carbonate content, and time of confinement on shear modulus and damping ratio at small and high shear strains are presented and then discussed. Relationships for the small-strain shear modulus, the degradation of shear modulus at high strains, and the increase of damping ratio at high strains over its small-strain value are proposed. Finally, the practical implications of the results in the context of seismic site response analysis are discussed.

Keywords Shear Modulus; Damping Ratio; Resonant-Column; Cohesive Soils; Seismic Response

1. Introduction

It is widely known in Soil Dynamics and Earthquake Engineering that deformation characteristics of soils, expressed in terms of shear stiffness, G , and damping ratio, DT , are the fundamental parameters describing soil behavior. These parameters are required for the estimation of the response of soil and soil-structure systems, when subjected to cyclic and dynamic loadings, i.e., earthquakes, vehicular traffic, machine vibration, pile-driving, blasting, etc. Moreover, the recognition of the importance of studying the stiffness properties below a strain level of 10^{-4} in static problems has shown — and it was verified by field measurements and experimental results — that dynamic and static properties should not any longer be distinguished from each other and that strain amplitude is the key parameter in determining soil behavior [Burland, 1989]. Therefore, accurate measurement of deformation characteristics at small (10^{-5}) to intermediate (10^{-5} – 10^{-3}) strains has become important in both dynamic and static problems over the past years.

Values of shear modulus and damping ratio at very small strains can be measured both in field and laboratory using dynamic techniques and through a wide range of strains in laboratory only. Resonant-column (RC) and combined torsional-shear/resonant-column tests are the most popular tests in determining shear modulus and damping ratio of soils.

This article presents results of RC tests, performed for the investigation of the dynamic properties of natural cohesive soils from Greece and Cyprus over the past ten years in the context of seismic site response and microzonation studies. The behavior of soils is described and the implications of the test results for seismic site response are discussed.

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2. Previous Laboratory Investigations on Shear Modulus and Damping Ratio

As stated previously, when a dynamic or static loading is imposed to soil, its response mainly depends on the level of strain induced to it. According to previous studies, soil behavior inside the limit state curve (LSC) ranges from linear elastic to inelastic and is divided in three zones; Fig. 1 [Jardine, 1992; Hight and Higgins, 1995]. At small strains (within zone 1, OA), the stress-strain relation is linear elastic, soil shear stiffness has its maximum value, $G = G_{\max}$, and damping ratio its minimum value, $DT = DT_{\min}$. The upper strain limit of this range is called linear elastic threshold shear strain, γ_t^e , and depends on soil type. When stress path crosses yield surface Y_1 , but remains inside zone 2 (between A and B), the stress-strain relation is nonlinear elastic, with some stiffness degradation and damping increase. The upper strain limit of this range is called

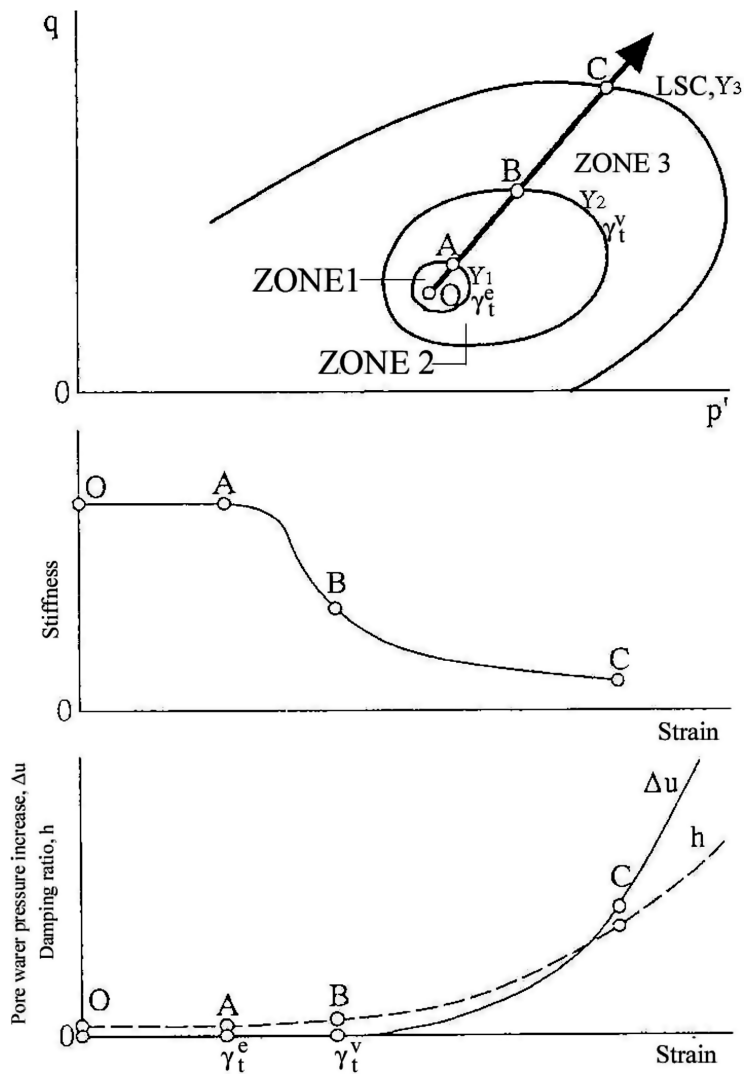


FIGURE 1 Schematic diagram showing the influence of strain on soil behavior [Jardine, 1992; Hight and Higgins, 1995].

volumetric threshold shear strain, γ_t^v , and depends on soil type and stress state. When stress path crosses surface Y_2 (between B and C in zone 3), the stress-strain behavior becomes inelastic with significant stiffness degradation, damping increase, and plastic strains development. Consequently, passage through curve Y_2 corresponds to γ_t^v , defined as the strain at which excess pore water pressures or plastic strains start to build up. A brief review of the main factors affecting dynamic properties, G and DT , of cohesive soils is presented below.

2.1. Small-Strain Shear Modulus and Damping Ratio

$G_{\max}$ is constant and independent of shear strain level for $\gamma \leq \gamma_t^e = 10^{-5}$. The main parameters affecting $G_{\max}$ are soil composition, void ratio, effective stress level, and structure. Jamiolkowski *et al.* [1985] and Hardin and Blandford [1989] proposed that, when a shear wave travels along i direction and is polarized along j direction, $G_{\max}$ on the plane normal to i direction may be expressed by the following general relationship:

$$G_{\max(ij)} = A_{ij} f(e) \sigma_i^{n_i} \sigma_j^{n_j} p_a^{(1-n_i-n_j)}, \quad (1)$$

where

- A_{ij} : dimensionless stiffness coefficient in plane ij ,
- $f(e)$: void ratio function,
- σ'_i, σ'_j : principal effective stresses,
- n_i, n_j : exponents of principal effective stresses, and
- p_a : reference pressure (equal to atmospheric pressure).

When shear modulus is measured in horizontal plane, as in RC tests, Eq. (1) becomes:

$$G_{\max(vh)} = A_{vh} f(e) \sigma_v^{n_v} \sigma_h^{n_h} p_a^{(1-n_v-n_h)}. \quad (2)$$

As shown by the above equations, $G_{\max}$ is a function of only two principal stresses, in the directions of wave propagation and particles movement, whereas it is not affected at all by the third principal stress, which is normal to propagation wave plane [Roesler, 1979]. Moreover, the equality of exponents ($n = n_v = n_h$), established by experimental results, allows the use of mean stress $\sigma'_m = (\sigma'_v + \sigma'_h)/2$ instead of σ'_v and σ'_h [Yu and Richart, 1984].

For isotropic stress conditions, Eq. (2) is further simplified to the following:

$$G_{\max(vh)} = A f(e) \sigma_o^n p_a^{1-n} \quad (3)$$

where

- σ'_o : isotropic effective stress and n : exponent of isotropic effective stress.

Table 1 summarizes the proposed relationships in literature for the estimation of $G_{\max}$ for cohesive soils, based on laboratory cyclic and dynamic tests. The proposed relationships for $G_{\max}$ have the following general form:

TABLE 1 Relationships for $G_{\max}$ of cohesive soils

General form of equation [†]	Constants & functions				Soil type	Testing method ^{††}	Reference
	A	f(e)	n	g(PI, OCR)			
$G_{\max} = A f(e) (\sigma'_o)^n$	3231	$(2.97-e)^2/(1+e)$	0.5		Kaolinite & Boston blue clay	R.C.	Hardin and Black [1968]
	445	$(4.4-e)^2/(1+e)$		0.5	Bentonite PI = 60%	R.C.	Marcuson and Wahls [1972]
	4500	$(2.97-e)^2/(1+e)$	0.5		Kaolinite PI = 25%		
	141	$(7.32-e)^2/(1+e)$	0.6		Thirty natural soils ($e = 1.5-4.0$ & PI = 40–100 %)	C.T.	Kokusho <i>et al.</i> [1982]
	9600	$1/(1 + 1.2e^2)$	0.5		Clays (PI = 9–27 % & $e = 0.583-1.068$)	R.C.	Vrettos and Savidis [1999]
$G_{\max} = A f(e) (\sigma'_v)^n$	5000	$1/e^{1.5}$	0.5		Seven clays	Lab & field tests	Shibuya and Tanaka [1996]
$G_{\max} = A f(e) (\sigma'_o)^n (p_a)^{(1-n)}$	370	$1/(0.3 + 0.7e)^2$	0.54		Silty clays	R.C.	Stokoe <i>et al.</i> [1995]
	600	$1/e^{1.3}$	0.5		Quaternary Italian clays	Lab tests	Jamiolkowski <i>et al.</i> [1995]
	358	$1/e^{1.21}$	0.57		Sandy silts – silty sands	R.C.	D'Elia and Lanzo [1996]
	506	$1/e^{1.1}$	0.42		Clayey silts		
$G_{\max} = A f(e) (\sigma'_o)^{ng} (OCR, PI)$	3231	$(2.97-e)^2/(1+e)$	0.5	OCR ^K , K = g(PI) [‡]	PI = 0–85 %	R.C.	Hardin and Black [1969]
	1576	$(2.97-e)^2/(1+e)$	0.5		$e = 0.48-1.36$ & PI = 12–30 %	R.C.	Kim and Novak [1981]

$G_{\max} = A f(e) (\sigma'_o)^n$ $\times (p_a)^{(1-n)g(OCR, PI)}$	625	$1/(0.3 + 0.7e)^2$	0.5	OCR ^K , K = g(PI) [‡]	Normally consolidated clays (e = 0.4–1.2)	R.C.	Hardin [1978]
$G_{\max} = A p_r^{(1-n)} (\sigma'_o)^n$ $\times g(OCR, PI)$	677	$1/(0.3 + 0.7e)^2$	0.487	OCR ^K , K = 0.419	Kaolinite	R.C.	Athanasopoulos [1981]
	100 + 2900 $\times (1-PI/50)$		$0.6 + 0.25 \times OCR^K$, (PI/50) < 0.85	K = 0.18 + 0.1 $\times (PI/50) < 0.28$	Clays p _r = 1kPa	B.E.	Viggiani and Atkinson [1995]
$G_{\max} = A g(OCR)$	47750			OCR ^K , K = 0.42	Kaolinite clay	R.C.	Athanasopoulos [1994]
$G_{\max} = (\sigma'_o)^{ng(PI)}$			1	285–(2 × PI)	PI > 30%	R.C.	Zen <i>et al.</i> [1987]
$G_{\max} = A (\sigma'_o)^{ng(PI)}$	280		0.82	$(1/(PI-Ir))^{(PI-Ir)}$ in decimal ¹	Mexico clay PI = 92–234% e = 4.00–9.37	R.C.	Romo [1995]
$G_{\max} = f(e) (\sigma'_o)^{ng(PI)}$		$1/(0.4 + 0.7e)$	1	358–(3.8 × PI)	PI = 30–50%	R.C.	Kagawa [1992]

[†]G_{max} & σ'_o in kPa and p_a: atmospheric pressure (= 100 kPa).

^{††}R.C.: Resonant-column test, C.T.: Cyclic triaxial test, B.E.: Bender element test.

[‡]K = 0, 0.18, 0.30, 0.41, 0.48, 0.50 for PI = 0, 20%, 40%, 60%, 80%, ≥ 100%, respectively.

¹Ir = (LL-w)/PI.

$$G_{\max} = Af(e) \sigma_o^n p_a^{1-n} g(OCR, PI). \quad (4)$$

$DT_{\min}$ is constant and independent of shear strain for $\gamma \leq \gamma_t^e = 10^{-5}$ and is controlled by viscosity of soil skeleton or pore liquid, depending on strain rate, $\dot{\gamma}$, or loading frequency, f , and soil type. According to Shibuya *et al.* [1995], for a given γ the variation of $DT_{\min}$ with $\dot{\gamma}$ or f is described by an U-shaped curve. For $f < 0.1$ Hz, $DT_{\min}$ increases with decreasing f , for $f = 0.1$ –10 Hz, $DT_{\min}$ remains constant with f , whereas for $f > 10$ Hz, $DT_{\min}$ increases with increasing f . Vinale *et al.* [2001] confirmed the U-shaped variation of $DT_{\min}$ with strain rate (or frequency) for a compacted silty sand ($PI = 13.7\%$). Vucetic and Dobry [1991] reported that $DT_{\min}$ ranges for most natural soils between approximately 0.5 and 5%. Vucetic *et al.* [1998] observed that for clayey soils at $\gamma < 5 \times 10^{-5}$, $DT_{\min}$ increases with increasing PI and is generally higher than sands. Higher values of $DT_{\min}$ have been determined for some special categories of soils, i.e., of the order of 8% for peats, 14% for soils with cemented bondings, and 9% for compacted soils [Kramer, 2000; Chang *et al.*, 2001; Tika *et al.*, 2005a]. This indicates that $DT_{\min}$ is influenced significantly by the structure of soils [Dobry, 1991].

2.2. Shear Modulus and Damping Ratio at High Strains

Vucetic and Dobry [1991] based on a review of cyclic loading test results for normally and overconsolidated clays ($OCR = 1$ –15), as well as sands, concluded that PI is the main factor controlling the variation of $G/G_{\max}$ and DT versus γ and proposed the curves shown in Fig. 2.

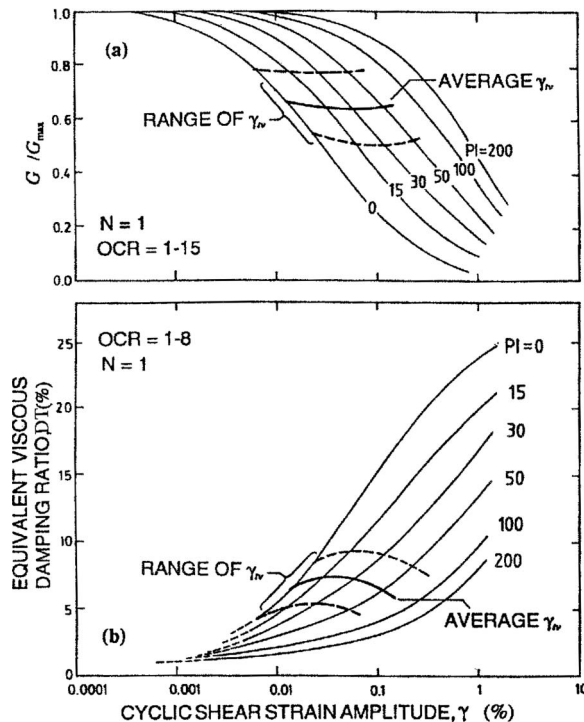


FIGURE 2 Influence of soil plasticity, PI , on variation of normalized shear modulus, $G/G_{\max}$, and damping ratio, DT , versus shear strain, γ , for normally and overconsolidated soils [Vucetic and Dobry, 1991; Vucetic, 1994].

As shown in the above figure at a given γ , higher values of PI correspond to higher values of $G/G_{\max}$ and lower values of DT. In contrast to small-strain range, where $DT_{\min}$ increases with increasing PI, at high-strain range DT decreases with increasing PI. This may be attributed to the change of damping mechanism with strain from viscous to hysteretic [Shibuya *et al.*, 1995].

A significant influence of consolidation stress, σ'_o , has only been observed for low-plasticity soils, where linearity of $G/G_{\max}$ - γ and DT - γ relationships increases with increasing σ'_o [Ishibashi and Zhang, 1993]. Void ratio, e , exerts a similar with PI but less strong influence on $G/G_{\max}$ - γ and DT - γ curves, since it depends on soil composition and structure, as well as consolidation stress and OCR [Hwang, 1997]. Other parameters that influence $G/G_{\max}$ and DT versus γ curves are number of loading cycles, N , and $\dot{\gamma}$. At high strains, the degradation effect of N on $G/G_{\max}$ - γ curves reduces with increasing PI [Idriss *et al.*, 1978]. On the contrary, G increases with increasing γ and this effect becomes more pronounced for soils of high plasticity at high strains [Lo Presti *et al.*, 1997]. At high strains, DT increases with increasing $\dot{\gamma}$ or loading frequency.

It is evident from the above that the threshold shear strains, γ_t^e and γ_t^v , referred to previously increase with increasing PI. Other parameters that influence both γ_t^e and γ_t^v are soil structure and strain rate, whereas N influences only γ_t^v [Jamiolkowski *et al.*, 1995].

Table 2 summarizes the proposed in literature relationships for the determination of $G/G_{\max}$ - γ and DT - $G/G_{\max}$ curves.

3. Apparatus and Testing Programme

3.1. Resonant-Column Apparatus

The tests have been performed in the RC apparatus, designed by Drnevich at University of Michigan [Drnevich, 1967]. The apparatus is of fixed-free type and allows both the longitudinal and torsional vibration of a solid cylindrical specimen (diameter 35.7/71.1 mm and height/diameter ratio 2:1). The specimen is placed in a triaxial cell, installed on a concrete base, which is fixed to the ground (passive end). It is surrounded by fluid (water) and subjected to an isotropic confining stress, σ_o , (up to 700 kPa) and back pressure, u_b . The dynamic excitation (force or moment) is subjected to the specimen by means of a system of magnets and coils, connected to the top base plate attached to the specimen (active end). The simple harmonic longitudinal or torsional vibration applied to the active end, produces longitudinal or shear waves, respectively, which propagate down to the specimen base (passive end) where reflect. The excitation frequency is changed until resonance in the first mode of vibration is achieved, and this occurs at a phase of 180° between excitation and velocity of active end. The resonance frequency, amplitude of vibration, and acceleration are measured at the active end of the specimen. These values together with the characteristics of the specimen (geometry and mass) and the apparatus (mass and stiffness of active end) are used for the estimation of propagating longitudinal and shear wave velocity, V_p and V_s , Young's and shear modulus, E and G , longitudinal and shear damping ratio, DL and DT , and axial and shear strain, ϵ and γ , respectively.

In this article, results for the torsional vibration are only presented. Calculations were made according to ASTM D4015–87. Considering the propagation of shear waves in a homogenous elastic rod with the particular boundary conditions of the resonant-column apparatus, the following equation is derived:

TABLE 2 Relationships for $G/G_{\max}\text{-}\gamma$ and DT- $G/G_{\max}$ curves of cohesive soils

General form of equation	Constants & functions			Reference
	A	m(PI), λ (PI)	n(γ , PI)-n _o	
$G_{\text{sec}}/G_{\text{max}}$ $= A(\gamma, PI)$ $\times \sigma_o^{m(\gamma, PI) - n_o}$	$0.5 \left\{ 1 + \tanh \left[\ln \left(\frac{0.000102 + m(PI)}{\gamma} \right)^{0.492} \right] \right\}$	$\begin{cases} 0.00 & PI = 0 \\ 3.37 \cdot 10^{-6} PI^{1.404} & 0 < PI \leq 15 \\ 7.00 \cdot 10^{-7} PI^{1.976} & 15 < PI \leq 70 \\ 2.70 \cdot 10^{-5} PI^{1.115} & PI > 70 \end{cases}$	$0.272 \left\{ 1 - \tanh \left[\ln \left(\frac{0.000556}{\gamma} \right)^{0.4} \right] \right\}$ $\times \exp(-0.0145 (PI)^{1.3})$	Ishibashi and Zhang [1993]
$DT = A$ $(G_{\text{sec}}/G_{\text{max}}, PI)$	$0.333 \frac{1 + \exp(-0.0145 PI^{1.3})}{2}$ $\times [0.586(G_{\text{sec}}/G_{\text{max}})^2 - 1.547(G_{\text{sec}}/G_{\text{max}}) + 1]$			
$G_{\text{sec}}/G_{\text{max}} = (PI)$	$1 - \left\{ \left(2700 \gamma \frac{G_{\text{sec}}}{G_{\text{max}}} \right)^{0.70} 10^{-(PI/\lambda)} \right\}$	$0.002(PI)^2 + 0.25(PI) + 60$		Michaelides <i>et al.</i> [1998]
$DT = A$ $(G_{\text{sec}}/G_{\text{max}}, PI)$	$\begin{cases} 2 + [18 - 0.08(PI - 15)] \\ \times (1 - G_{\text{sec}}/G_{\text{max}}) & 0 \leq PI \leq 100 \\ 2 + 11.2(1 - G_{\text{sec}}/G_{\text{max}}) & PI > 100 \end{cases}$			

$$\frac{I}{I_o} = \left(\frac{2 \pi f_n l}{V_s} \right) \tan \left(\frac{2 \pi f_n l}{V_s} \right), \quad (5)$$

where

- I : polar moment of inertia of soil specimen,
- I_o : polar moment of inertia of top platen mass
at the active (free) end of soil specimen,
- f_n : resonance frequency and
- l : specimen length.

The shear modulus is then evaluated as follows:

$$G = \rho V_s^2 \quad (6)$$

where

ρ : soil density.

3.2. Tested Soils

The tested soils were retrieved from various sites of Greece and Cyprus, using either rotary core, or thin wall tube sampling (shelby plastic or metal). The soils which were tested in an undisturbed state are classified into the following three groups according to their calcium carbonate content, CC, defined as percentage of dry soil sample mass:

1. Group A: soils with CC < 5%, described as silts, clays, and clayey sands
2. Group B: soils with CC = 5–16 %, described as marly clays/clayey-sands
3. Group C: soils with CC = 26–86 %, described as clayey marls, marls, and marly limestone

Tables 3–5 present the physical and mechanical properties of the tested undisturbed soils.

In addition, a group of reconstituted samples (R), which were formed in the laboratory from natural soils and kaolinite (4R), were tested. The samples were prepared from a slurry at water content ranging from 1.0 and 1.5 times the liquid limit, LL, of the soils and were then consolidated under one-dimensional conditions up to a vertical stress of 100 kPa. Table 6 presents the physical properties of the reconstituted soils.

3.3. Testing Procedure

Each specimen, after it had been trimmed to its nominal dimensions, was installed in the triaxial cell base and then enclosed by a membrane. The apparatus was assembled and saturation of the specimen followed by increasing both the back and cell pressure, but keeping a constant effective stress ($\sigma'_o \approx 30$ kPa). Saturation degree was evaluated by estimating Skempton's parameter B . After the completion of saturation procedure, the specimen was consolidated at various levels of σ'_o . Length and volume changes of the specimen are continuously measured. Two types of dynamic excitation were imposed: small-strain loading and a sequence of loadings with increasing strains. The former was applied sometimes during consolidation stage and always after the completion of primary consolidation, whereas the latter was applied always after the completion of primary consolidation. As the test is considered to be non destructive, σ'_o was usually increased to the next level, in order to assess the dependence of dynamic properties on σ'_o .

TABLE 3 Physical and mechanical properties of the undisturbed tested soils with CC < 5% (Group A)

Soil no. [†]	σ'_v ^{††} (kPa)	w _o (%)	LL (%)	PI (%)	CF (%) (<2 μ m)	γ (kN/m ³)	G _s	e _o	Sr _o (%)	OCR	Sampling Type [‡]	USCS	Reference
1A	145	15	25	5	7	21.1	2.625	0.397	96	1.7 [†]	R	CL-ML	Tika [1996]
2A	435	15	21	5	5	21.5	2.712	0.420	94	1.7 [†]	PL	SC-SM	»
3A	180	22	25	6	23	21.2	2.808	0.590	100	1.0	PL	CL-ML	Kallioglou [2003]
4A	97	19	29	6	20	20.9	2.840	0.586	91	2.0	PL	ML	»
5A	116	21	32	8	24	20.8	2.810	0.600	96	1.6 [†]	R	ML	»
6A	103	23	28	8	15	20.2	2.700	0.617	100	1.7 [†]	PL	CL	Tika [1996]
7A	311	12	28	9	17	21.4	2.674	0.365	84	4.0 ^{!!}	R	CL-SC	Kallioglou [2003]
8A	566	18	29	10	26	21.5	2.805	0.509	99	1.0	R	CL	»
9A	273	16	28	12	32	21.1	2.669	0.444	99	5.0 ^{!!}	R	CL	Tika [1996]
10A	588	18	35	12	9	20.5	2.700	0.531	95	1.7 [†]	R	CL	»
11A	423	12	30	13	22	21.5	2.711	0.389	85	2.0 ^{!!}	R	CL-SC	Kallioglou [2003]
12A	169	16	33	14	24	20.9	2.696	0.476	93	4.0 ^{!!}	R	CL	»
13A	185	14	31	14	19	21.6	2.764	0.424	88	12.2 ^{!!}	PL	CL	»
14A	733	27	42	18	9	19.0	2.700	0.769	95	1.7 [†]	R	CL	Tika [1996]
15A	73	16	39	18	14	20.3	2.656	0.497	88	3.3 [†]	R	CL	Kallioglou [2003]
16As	147	21	37	19	29	20.3	2.768	0.620	94	1.5	PL	CL	»
17A	171	14	39	21	24	21.6	2.757	0.434	90	7.0 ^{!!}	R	CL	Tika [1996]
18As	50	27	56	30	46	19.3	2.683	0.731	97	3.0	PL	CH	Kallioglou [2003]
19A	783	28	58	37	13	19.4	2.642	0.707	100	1.7 [†]	R	CH	Tika [1996]

[†]Specimens had nominal dimensions: D_o = 7.1 cm, H_o = 14.2 cm, apart from those indicated with index s (16As & 18As), which had nominal dimensions: D_o = 3.6 cm, H_o = 7.1 cm.

^{††}Estimated in situ vertical effective stress. Where data were not available, the unit weight, γ , above and below ground water table assumed equal to 18kN/m³ and 20kN/m³, respectively. Unit weight of water, γ_w , assumed equal to 10kN/m³.

[†]Assumed using one-dimension consolidation test results of nearby samples within the same soil layer.

^{!!}Estimated using the equation proposed by Jamiolkowski *et al.* [1985] $c_u/\sigma'_v = 0.27 \times OCR^{0.8}$. When c_u values were not available, Stroud's relationship between N_{SPT} and c_u for cohesive soils was used [1974]. The effect of σ'_v in N_{SPT} was taken into account.

[‡]Sampling using thin wall plastic tube (PL) or rotary core (R).

TABLE 4 Physical and mechanical properties of the undisturbed tested soils with $5\% \leq CC \leq 16\%$ (Group B)

Soil no. [†]	σ'_v ^{††} (kPa)	w _o (%)	LL (%)	PI (%)	CF (%)		G _s	e _o	Sr _o (%)	CC ^h (%)	OCR	Sampling Type [‡]	USCS	Reference
					(<2μm)	γ (kN/m ³)								
1B	131	18	30	9	28	20.7	2.774	0.547	90	(<15) [*]	5.7 ^{!!}	R	CL	Tika [1996]
2B	119	20	31	11	14	19.8	2.683	0.596	90	(<14) [*]	10.0 ^{!!}	R	SC	Tika [1996]
3B	163	21	29	11	16	20.6	2.710	0.527	100	9–6	2.0	PL	CL	Kallioglou [2003]
4B	637	17	33	12	27	21.4	2.790	0.495	96	6	1.0	R	CL	»
5B	532	23	36	13	13	20.6	2.698	0.585	100	(< 7) [*]	1.0 [!]	R	CL	»
6B	201	17	29	13	17	21.5	2.723	0.449	100	9–16	1.0	PL	CL	»
7B	469	15	38	14	8	22.0	2.732	0.397	100	7	1.0	PL	SC	»
8B	154	21	35	19	25	20.5	2.631	0.512	100	7–6	1.8 [!]	PL	CL	»
9B	246	13	40	19	7	21.8	2.709	0.380	96	12–11	11.0 ^{!!}	PL	SC	»
10B	124	20	40	20	32	21.3	2.900	0.601	93	5–6	1.0	R	CL	»
11B	121	26	44	24	32	18.9	2.719	0.771	90	7–3	1.6	M	CL	»
12B	71	14	52	29	14	20.7	2.715	0.471	83	14	15.0 ^{!!}	R	CH	»

[†]Specimens had nominal dimensions: D_o = 7.1 cm, H_o = 14.2 cm.

^{††}Estimated insitu vertical effective stress. Where data were not available, the unit weight, γ, above and below ground water table assumed equal to 18kN/m³ and 20kN/m³, respectively. Unit weight of water, γ_w, assumed equal to 10kN/m³.

^hCalcium carbonate content values determined using HCl method (first value) and radiographic method (second value). For soils 3B, 6B, 8B, 9B, and 10B NaOAC-HOAC method (first value) was used.

^{*}Calcium carbonate content assumed to be the same of that of nearby samples within the same soil layer.

[!]Assumed using one-dimension consolidation test results of nearby samples within the same soil layer.

^{!!}Estimated using equation proposed by Jamiolkowski *et al.* [1985] $c_u/\sigma'_v = 0.27 \times OCR^{0.8}$. When c_u values were not available, Stroud's relationship between N_{SPT} and c_u for cohesive soils was used [1974]. The effect of σ'_v in N_{SPT} was taken into account.

[‡]Sampling using thin wall plastic tube (PL), thin wall metal tube (M), or rotary core (R).

TABLE 5 Physical and mechanical properties of the undisturbed tested soils with $26\% \leq C \leq 86\%$ (Group C)

Soil no. [†]	σ'_v ^{††} (kPa)	w_o (%)	LL (%)	PI (%)	CF (%) ($<2\mu\text{m}$)	γ (kN/m ³)	Gs	e_o	Sr _o (%)	CC ^h (%)	OCR	Sampling Type [‡]	USCS	Reference
1C	53	17	32	7	16	19.1	2.696	0.620	73	61	1.9 ^{!!}	R	ML	Kallioglou [2003]
2C	93	29	34	10	18	19.3	2.732	0.785	99	42	1.0 ^{!!}	R	CL	Kallioglou [2003]
3C	123	33	39	12	24	18.3	2.759	0.965	93	78–86	2.0	PL	ML	»
4C	214	26	33	12	25	19.7	2.752	0.723	97	(>35) [*]	1.5 [!]	R	CL	Tika <i>et al.</i> [2005a]
5C	288	21	32	13	47	20.2	2.745	0.614	93	(>35) [*]	1.0 [!]	R	CL	Tika [1996]
6C	155	26	36	14	25	19.4	2.694	0.745	94	(>35) [*]	2.0 [!]	PL	CL	Tika <i>et al.</i> [2005a]
7C	97	27	35	16	26	19.6	2.758	0.751	97	(>35) [*]	2.0 [!]	PL	CL	»
8C	168	28	44	18	4	19.6	2.795	0.780	99	31	1.0 ^{!!}	PL	CL-ML	Kallioglou [2003]
9C	233	18	43	21	38	21.1	2.741	0.504	98	(>35) [*]	4.0 ^{!!}	R	CL	Tika <i>et al.</i> [2005a]
10C	137	31	49	24	25	18.7	2.711	0.870	97	52–67	1.0 ^{!!}	R	CL	Kallioglou [2003]
11C	66	37	57	25	11	18.2	2.761	1.037	98	(26–45) [*]	1.3 ^{!!}	R	MH-CH	»
12C	241	36	56	26	30	18.2	2.788	1.046	96	26–45	2.2	PL	CH-MH	»
13C	174	25	52	27	31	19.7	2.747	0.719	99	31	9.2	R	CH	Tika <i>et al.</i> [2005a]
14C	257	26	52	27	45	19.6	2.706	0.715	100	53	5.5	R	CH	»
15C	772	31	51	27	45	18.9	2.735	0.858	99	57	2.6	R	CH	»
16C	230	28	50	28	37	118.5	2.698	0.825	91	(42–65) [*]	1.0 ^{!!}	R	CH	Kallioglou [2003]
17C	35	42	59	28	50	17.9	2.725	1.118	100	28	2.5 [!]	PL	CH-MH	»
18C	108	23	46	28	31	20.2	2.795	0.673	95	(>35) [*]	3.0 [!]	R	CL	Tika [1996]
19C	157	28	52	29	33	19.5	2.773	0.779	96	34	2.9	R	CH	Tika <i>et al.</i> [2005a]

20C	462	31	52	29	40	19.0	2.745	0.850	99	51	3.8 ^{!!}	R	CH	Kallioglou [2003]
21C	92	42	57	29	23	17.5	2.752	1.181	97	35–45	5.4	PL	CH-MH	»
22Cs	74	23	55	29	59	19.9	2.547	0.545	100	46–69	3.5 ^{!!}	M	CH	»
23C	77	24	51	30	37	19.8	2.798	0.717	91	(>35)*	4.5 [!]	PL	CH	Tika [1996]
24C	180	38	58	32	51	18.1	2.709	1.024	100	52	11.1	R	CH	Tika <i>et al.</i> [2005a]
25C	308	47	63	34	43	17.1	2.809	1.362	96	32–36	1.5	R	CH	Kallioglou [2003]
26C	219	33	63	34	35	18.9	2.717	0.872	100	40	7.8	R	CH	Tika <i>et al.</i> [2005a]
27C	64	35	61	37	34	18.2	2.712	0.971	98	(>35)*	2.0 ^{!!}	R	CH	Tika [1996]
28C	235	34	68	39	51	18.5	2.735	0.948	99	34	3.7	R	CH	Tika <i>et al.</i> [2005a]
29C	155	57	108	47	30	15.2	2.645	1.689	89	41	1.5	R	MH	Kallioglou [2003]
30C	189	43	99	66	73	18.6	2.682	1.017	100	(>35)*	2.1 [!]	R	CH	»

[†]Specimens had nominal dimensions: $D_o = 7.1$ cm, $H_o = 14.2$ cm, apart from that indicated with index s (22Cs), which had nominal dimensions: $D_o = 3.6$ cm, $H_o = 7.1$ cm.

^{††}Estimated insitu vertical effective stress. Where data were not available, the unit weight, γ , above and below ground water table assumed equal to 18kN/m^3 and 20kN/m^3 , respectively. Unit weight of water, γ_w , assumed equal to 10kN/m^3 .

^hCalcium carbonate content values determined using HCl method (first value) and radiographic method (second value).

*Calcium carbonate content assumed to be the same of that of nearby samples within the same soil layer.

[!]Assumed using one-dimension consolidation test results of nearby samples within the same soil layer.

^{!!}Estimated using the equation $c_u/\sigma'_v = 0.672 \times \text{OCR}^{0.831}$, proposed for marls tested in AUTH [Tika *et al.*, 2005b]. When c_u values were not available, Stroud's relationship between N_{SPT} and c_u for cohesive soils was used [1974]. The effect of σ'_v in N_{SPT} was taken into account.

[‡]Sampling using thin wall plastic tube (PL), thin wall metal tube (M), or rotary core (R).

TABLE 6 Physical properties of the reconstituted tested soils

Soil no. [†]	w _o ^{††} (%)	LL (%)	PI (%)	CF (%) (<2μm)	Gravel (%)		γ (kN/m ³)	Gs	e _o	Sr _o (%)	CC ^h (%)	USCS	Reference
					(>2mm)								
1R	23	32	10	11	0	21.3 ^{!!}	2.727	0.550 ^{!!}	100	0	CL	Tika <i>et al.</i> [2005a]	
2R	33	44	18	21	0	19.5 ^{!!}	2.780	0.852 ^{!!}	100	0	CL		»
3R	45	53	25	34	0	17.1 ^{!!}	2.674	1.221 ^{!!}	91	15	CH		»
4R	49	54	28	48	0	16.9 ^{!!}	2.692	1.331 ^{!!}	99	0	CH		»
5R	55	70	43	58	0	16.6 ^{!!}	2.768	1.525 ^{!!}	100	0	CH		»

[†]Specimens had nominal dimensions: D_o = 7.1 cm, H_o = 14.2 cm.
^{††}The values refer to the reconstituted specimens after their consolidation (at σ_v' = 100kPa) from a slurry with w>1.0LL.
^hCalcium carbonate content values determined using HCl method.

4. Test Results & Discussion

It is well known that natural soils are structured materials. As stated by Mitchel [1976], the term “structure” means the combination of “fabric” (particle arrangement) and interparticle “bonding”. The structure of natural clays depends on many factors, such as depositional conditions, aging, cementation, and leaching. Burland [1990] proposed the use of a reference state in terms of the properties of reconstituted clays (consolidated at water contents between 1.1 LL and 1.5 LL), with which those of undisturbed clays are compared. The properties of the reconstituted clays are termed “intrinsic” properties since they are inherent to the soil and independent of their natural state. For the undisturbed tested soils in this work, the presence of calcium carbonate, which may act as cementing agent at particle contents, indicates possibly the existence of a bonded structure and thus an extra component of strength and stiffness. It may be considered, therefore, that the structure of soils of group A arises from soil fabric, whereas that of soils of groups B and C may also be due to cementation.

4.1. Small-Strain Shear Modulus

In order to assess the effect of structure on dynamic properties of undisturbed soils, the results of the reconstituted and then of the undisturbed soils are presented and discussed.

The variation of $G_{\max}$ at 24 h confinement time with σ'_o for reconstituted soils is shown in Fig. 3. A linear relationship between $G_{\max}$ and σ'_o in a log-log plot is observed for each soil. The different relationships indicate the effect of PI and e , parameters describing the composition and state of soils, respectively. The intrinsic $G_{\max}$ can be expressed by the following equation:

$$G_{\max} = (5660 - 80PI) \sigma_o^{0.5} e^{-0.63} \quad R^2 = 0.930 \quad (7)$$

where

$G_{\max}$ and σ'_o are expressed in kPa and PI as a percentage value (%).

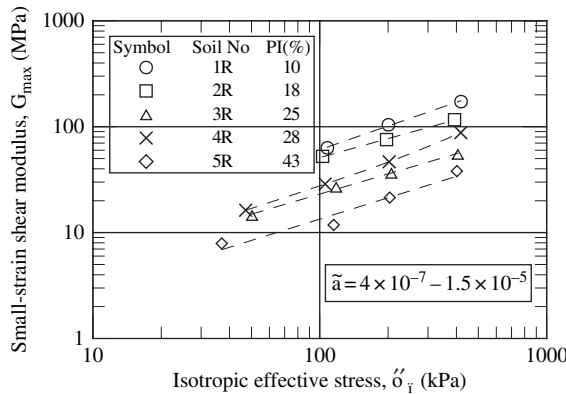


FIGURE 3 Variation of small-strain shear modulus, $G_{\max}$, with isotropic effective stress, σ'_o , for reconstituted soils.

The variation of $G_{\max}$ at 24 h confinement time with σ'_o for the undisturbed soils of the three groups is shown in Figs. 4–6, for PI lower and higher than 15% separately. Similarly with the reconstituted soils, a linear relationship between $G_{\max}$ and σ'_o in a log-log plot is observed for each of the above soils at both normal consolidation and over-consolidation states. The different relationships indicate the effect of void ratio, plasticity, and structure of the soils. For the soils of group A obtained by thin wall tube sampling, as they are considered to be of higher quality, $G_{\max}$ at 24 h confinement time, at stresses equal or higher than the estimated insitu mean effective stress, σ_o^{insitu} , can be expressed by the following equation:

$$G_{\max} = (6290 - 80PI) \sigma_o^{0.5} e^{-0.63} \quad R^2 = 0.793 \tag{8}$$

where

$G_{\max}$ and σ'_o are expressed in kPa and PI as a percentage value (%).

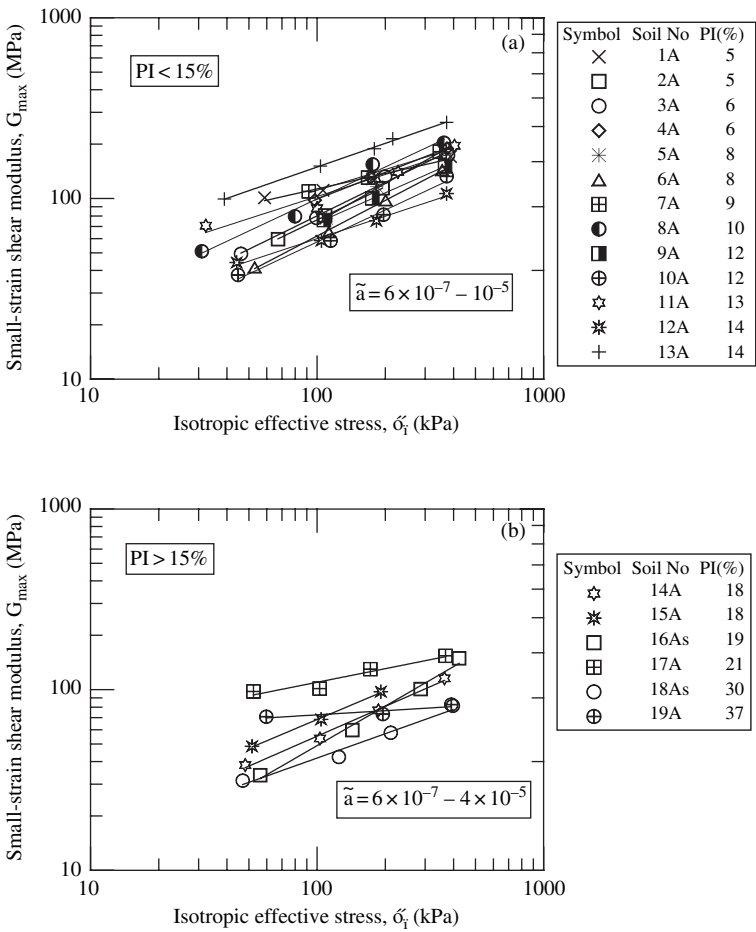


FIGURE 4 Variation of small-strain shear modulus, $G_{\max}$, with isotropic effective stress, σ'_o , for undisturbed soils with $CC < 5\%$.

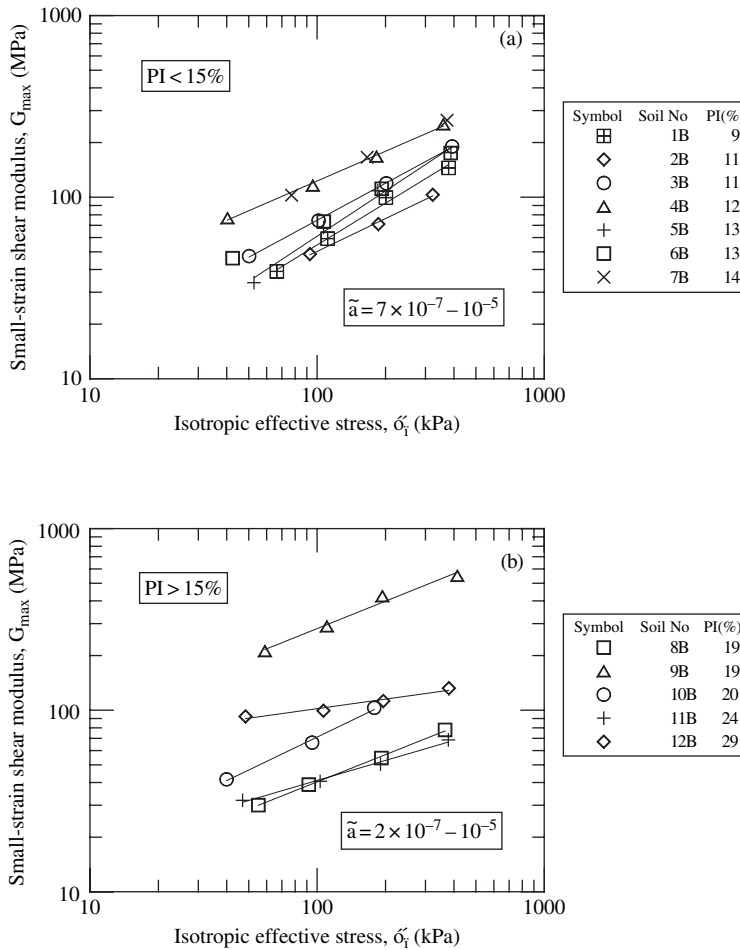


FIGURE 5 Variation of small-strain shear modulus, $G_{\max}$, with isotropic effective stress, σ'_o , for undisturbed soils with $5\% \leq CC \leq 16\%$.

The insitu mean effective stress, $\sigma'_o{}^{\text{insitu}}$, was determined using the value of the coefficient of earth pressure at rest, k_o , estimated either using Jaky's [1944] equation for normally consolidated clays ($1 - \sin \phi'$), or as function of PI and OCR for overconsolidated clays [Brooker and Ireland, 1965]. The angle of shearing resistance, ϕ' , was evaluated either from the empirical correlation of ϕ' and PI, or from triaxial test results, when such results were available.

Figure 7 summarizes the variation of $\frac{G_{\max}}{f(e)g(PI)}$ values using Eq. (8) at in situ stress level, $\sigma'_o{}^{\text{insitu}}$, for all undisturbed, obtained by both thin wall tube and rotary core sampling, as well as reconstituted soils. As indicated by comparing Eqs. (7) and (8) and also shown in Fig. 7, reconstituted soils exhibit lower values of normalized $G_{\max}$ than soils of group A. This difference may be attributed to the aging, as well as soil fabric developed in latter soils at their natural state. Soils of group B have similar values of normalized $G_{\max}$ with soils of group A and soils of group C show either similar or increased values of normalized $G_{\max}$. Values of normalized $G_{\max}$ similar with those of group A indicate that the stiffness component due to cementation ($CC \geq 5\%$) is either

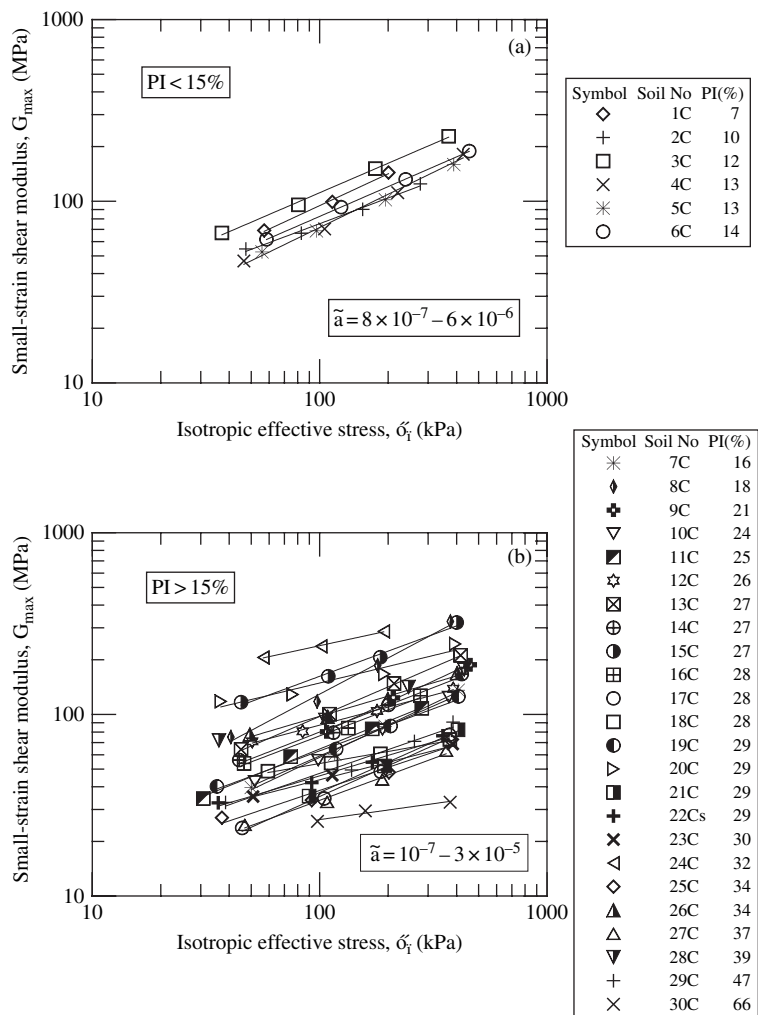


FIGURE 6 Variation of small-strain shear modulus, $G_{\max}$, with isotropic effective stress, σ'_o , for undisturbed soils with $26\% \leq CC \leq 86\%$.

insignificant, or lost during the sampling process, whereas higher values of normalized $G_{\max}$ imply that strong interparticle bonding had been developed. It may be anticipated that soils with calcium carbonate content from 17–25% have normalized $G_{\max}$ values similar with that of soils of group B.

The effect of time on $G_{\max}$ was studied by conducting tests of long duration under constant consolidation stress and recording the variation of $G_{\max}$ with secondary consolidation time. A linear increase of $G_{\max}$ with logarithm of time was observed. The influence of secondary compression time on $G_{\max}$ has been quantified in terms of the following parameter N_G [Marcuson and Wahls, 1972]:

$$N_G = \frac{G_t - G_{t_p+1440}}{\log\left(\frac{t}{t_p+1440}\right) G_{t_p+1440}} \quad (9)$$

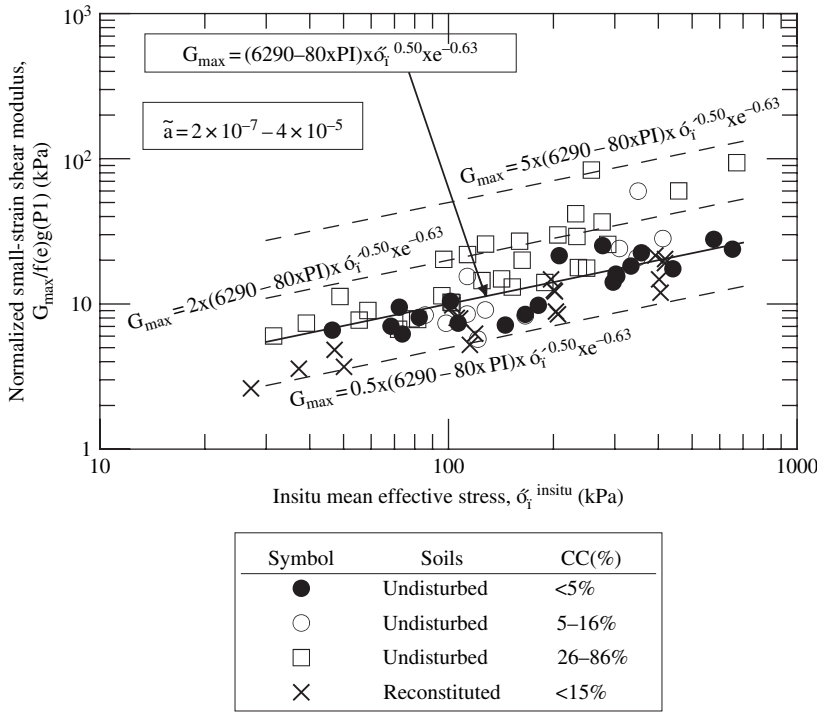


FIGURE 7 Variation of normalized small-strain shear modulus, $G_{\max}/f(e)g(PI)$, with insitu mean effective stress, $\sigma'_o{}^{\text{insitu}}$, for all undisturbed and reconstituted soils.

where

G_t : small – strain shear modulus at a given time $t(\text{min})$

G_{t_p+1440} : small – strain shear modulus at 24 h (1,440 min) after the end of primary consolidation $t_p(\text{min})$.

As shown in Fig. 8, where results for undisturbed soils at stresses equal or higher than $\sigma'_o{}^{\text{insitu}}$ and reconstituted soils at all stresses are presented, N_G increases with increasing PI. It is also shown that the values of N_G for reconstituted soils are higher than the corresponding values for undisturbed soils at the same plasticity, possibly due to lack of aging history. For the undisturbed soils, N_G increases with decreasing σ'_o .

Figure 9 shows a comparison of small-strain shear velocity, V_s , obtained by the RC tests of this work and field seismic tests (cross hole and down hole) at exactly the same field points (borehole and depth). The difference between the RC and the field seismic test results may be attributed to sampling disturbance, different boundary conditions and also applied stresses. Soils with largest discrepancies were those for which the rotary core sampling was used. Part of the sampling induced disturbance may be recovered by increasing the duration of secondary compression time in RC tests.

4.2. Small-Strain Damping Ratio

The variation of $DT_{\min}$ at 24 h confinement time with σ'_o for reconstituted soils is shown in Fig. 10a. A rather small change of $DT_{\min}$ with increasing σ'_o is indicated. The values of

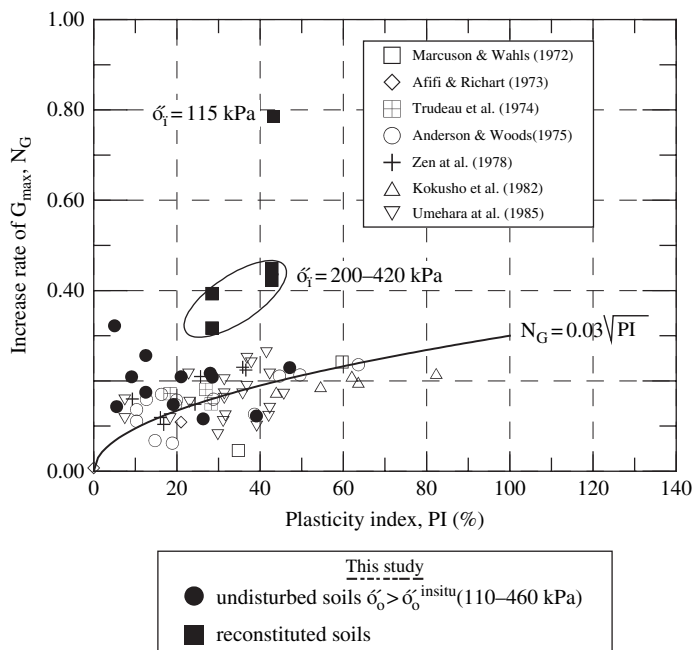


FIGURE 8 Variation of N_G with plasticity index, PI.

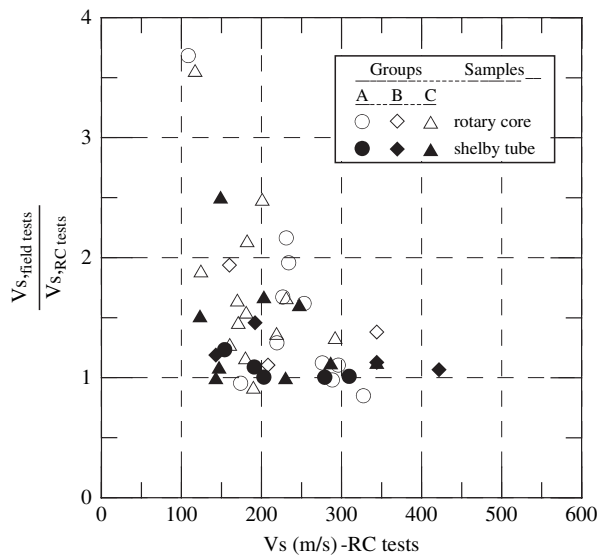


FIGURE 9 Comparison of small-strain shear wave velocity values from RC tests and field seismic tests [Pitilakis *et al.*, 1998; 1999; Tika *et al.*, 2001].

DT_{min} for the tested reconstituted soils vary between 1.2 and 3.4%. A comparison of DT_{min} of reconstituted soils 1R and 2R and that of same soils tested in a saturated compacted state is also given in the above figure. The higher values of DT_{min} (5.4–7.4%) for the latter samples confirm the effect of structure on DT_{min} . By considering the

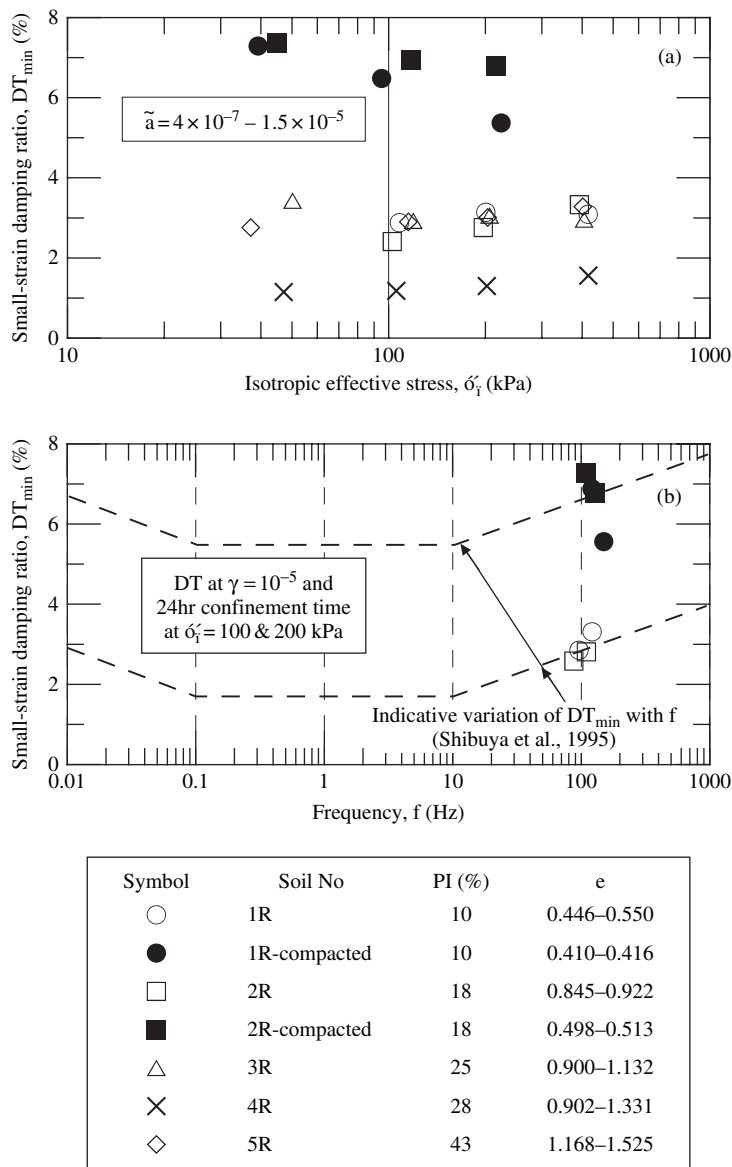


FIGURE 10 Effect of (a) isotropic effective stress, σ'_o , and (b) resonance frequency, f_n , on small-strain damping ratio, DT_{min} , for reconstituted soils (open symbols) and saturated compacted soils (closed symbols) [Tika *et al.*, 2005a].

U-shaped variation of DT_{min} with loading frequency, mentioned previously, and the range of resonance frequency, encountered in the RC tests, it may be concluded that the structure of the soils in the compacted state is more sensitive to strain-rate or loading frequency than that of the same soils in the reconstituted state; see Fig. 10b.

The variation of DT_{min} at 24 h confinement time with σ'_o for undisturbed soils is shown in Fig. 11. DT_{min} remains either constant or decreases with σ'_o , for both normal consolidation and overconsolidation states. The results also show that contrary to previous studies, e has a stronger influence on DT_{min} than PI [Vucetic *et al.*, 1998].

Figure 12 presents the variation of DT_{min} at σ_o^{insitu} and 24 h confinement time with void ratio for all the undisturbed soils tested. DT_{min} ranges between 1.4 and 4.7% for soils with e values greater than 0.5. However, for soils having lower void ratios ($e < 0.5$), DT_{min} varies from 2.2–8.7%. The test results presented here define a broader range of values of DT_{min} than that reported previously between 0.5 and 5% [Vucetic and Dobry, 1991]. By the same reasoning given above, when comparing the DT_{min} values of reconstituted and compacted soils, it may be concluded that the structure of some of the soils with low void ratios ($e < 0.5$) may be more sensitive to strain-rate or loading frequency than others with similar or higher void ratios. These high values of DT_{min} may be attributed to the

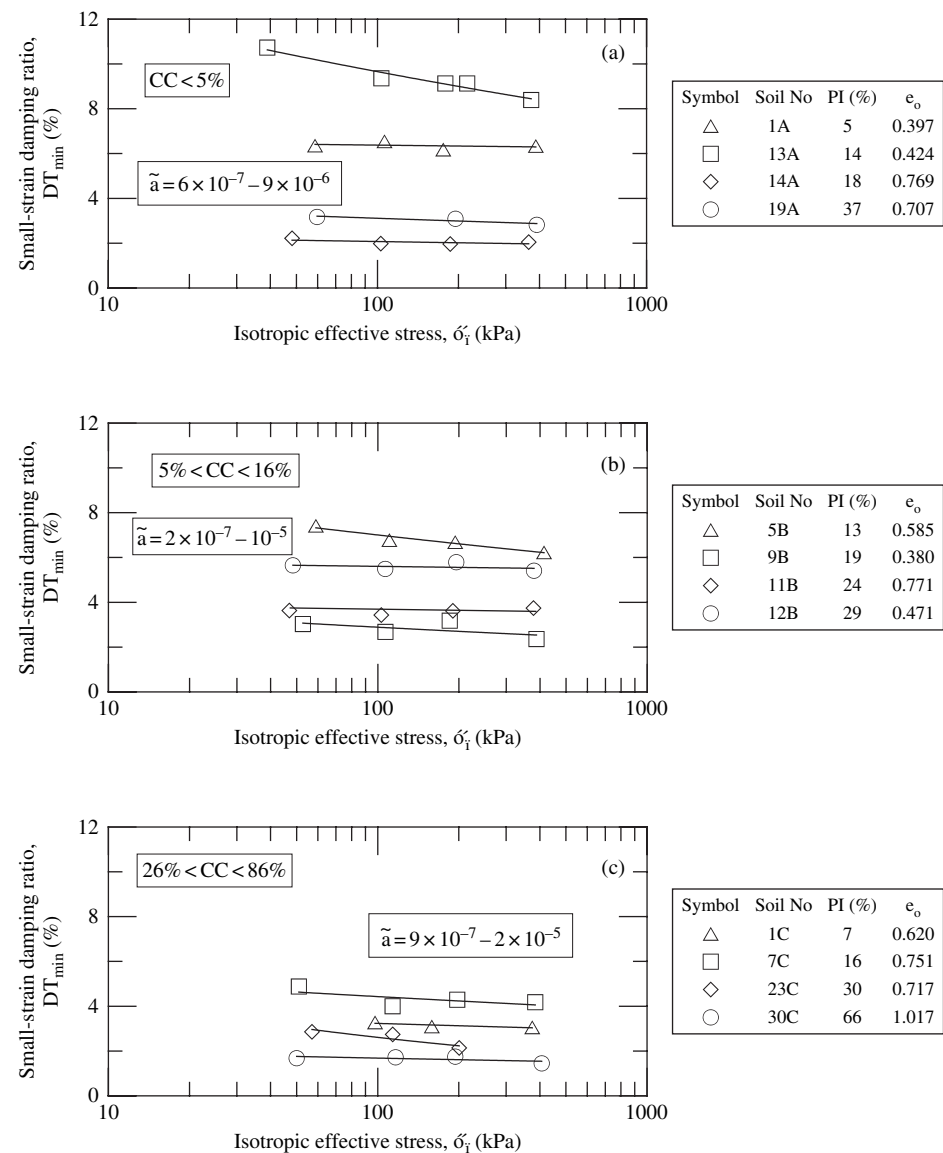


FIGURE 11 Variation of small-strain damping ratio, DT_{min} , with isotropic effective stress, σ'_o , for undisturbed soils.

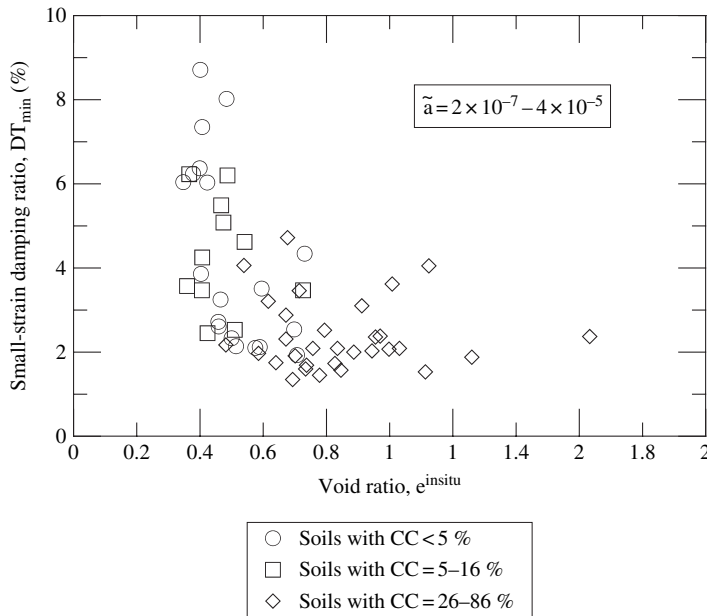


FIGURE 12 Variation of small-strain damping ratio, DT_{min} , with void ratio at insitu mean effective stress, e^{insitu} , for all undisturbed soils.

increased interaction of soil particles and thus an increased dissipation of energy at particle contacts.

In contrast with G_{max} , the influence of secondary consolidation time on DT_{min} values was found to be insignificant.

Although there are limited data, a good agreement of DT_{min} values determined by RC tests and field seismic tests, specifically the Down-Hole (DH) and Spectral Analyses of Surface Wave (SASW) methods, has been observed; see Table 7.

4.3. High-Strain Shear Modulus and Damping Ratio

The effect of shear strain, γ , on normalized shear modulus, G/G_{max} , and damping ratio, DT , is shown in Fig. 13 for the reconstituted soils at various levels of σ'_o and in Figs. 14–16

TABLE 7 Comparison of DT_{min} determined by RC and field seismic tests

Soil no.	PI (%)	DT_{min} -RC (%) [†]	DT_{min} -DH (%) ^{††}	DT_{min} -SWAS (%) ^{††}
6A	8	2.1	2.4	2.4
2B	11	1.9	2.3	
3C	12	6.1	6.1	
14A	18	1.9	1.6	1.6
9B	19	6.2	8.2	8.5–12.5
27C	37	2.4	2.8	

[†]Determined by the RC tests presented in this work.

^{††}Raptakis *et al.* [1997], Pitilakis *et al.* [1999] and Tika *et al.* [2001].

for the undisturbed soils at $\sigma_o^{\prime insitu}$ approximately. The degradation curves of shear modulus for both reconstituted and undisturbed soils follow the trend described in previous studies, with the G/G_{max} versus γ relationship shifting to higher values of γ with increasing PI [Vucetic and Dobry, 1991; Sun *et al.*, 1988], whereas damping ratio variation curves exhibit significant differences from the proposed in literature curves.

The effect of $\sigma_o^{\prime}$ on G/G_{max} - γ and DT - γ curves for reconstituted soils, as presented in Fig. 13, is negligible. Similarly, the G/G_{max} - γ and DT - γ curves for undisturbed soils tested at various levels of $\sigma_o^{\prime}$, higher than $\sigma_o^{\prime insitu}$ for each soil, are

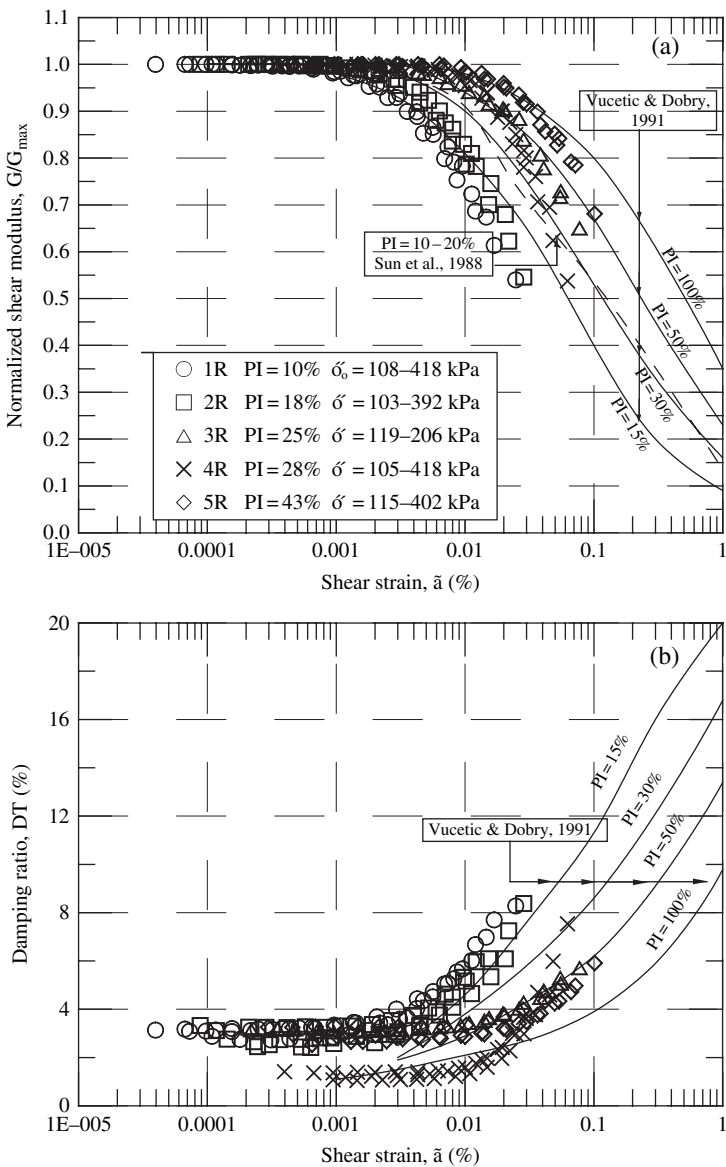


FIGURE 13 Variation of (a) normalized shear modulus, G/G_{max} , and (b) damping ratio, DT , with shear strain, γ , at various levels of $\sigma_o^{\prime}$ for reconstituted soils.

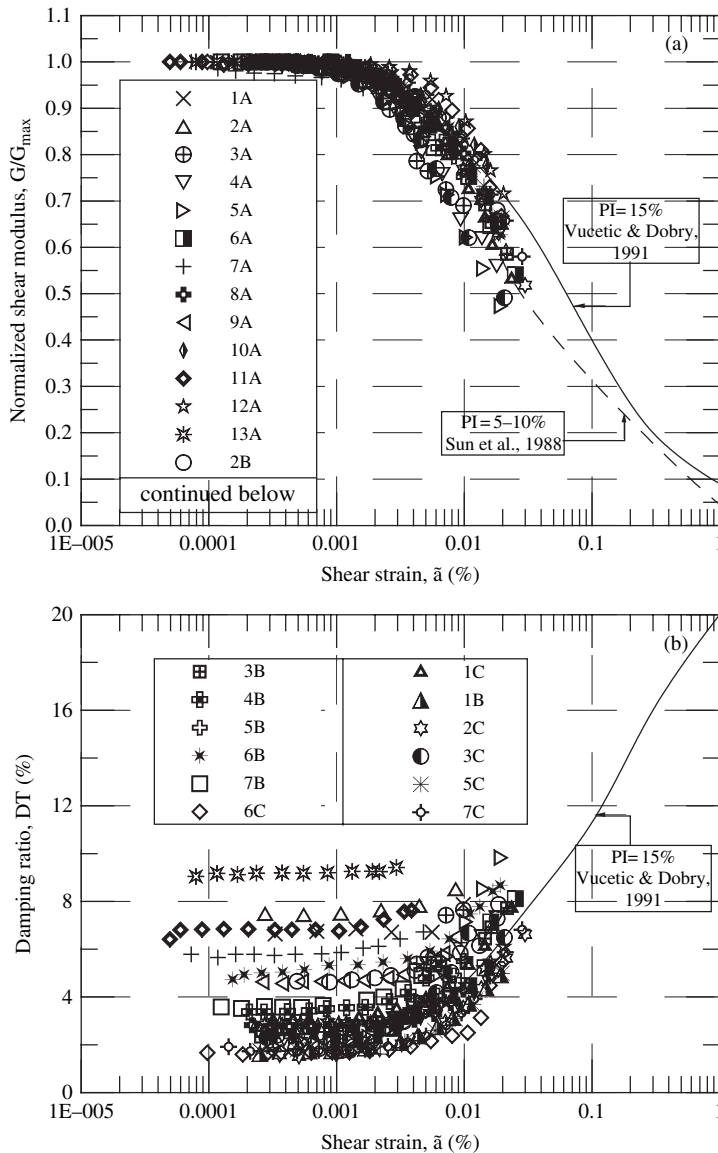


FIGURE 14 Variation of (a) normalized shear modulus, $G/G_{\max}$, and (b) damping ratio, DT , with shear strain, γ , at $\sigma'_o{}^{\text{insitu}}$ for undisturbed soils with plasticity index, $PI = 5-15\%$.

independent of σ'_o . A negligible effect of interparticle bonding, as indicated by calcium carbonate content, on $G/G_{\max}$ - γ curves was also observed; see Fig. 17a. The difference in DT - γ curves indicated in Fig. 17b is mainly due to the different values of $DT_{\min}$ of the soils.

The effect of PI on the degradation curves of shear modulus for all tested soils is shown in a 3D diagram; see Fig. 18. The data allow the determination of the following equation for the normalized shear modulus, $G/G_{\max}$, as a function of γ and PI :

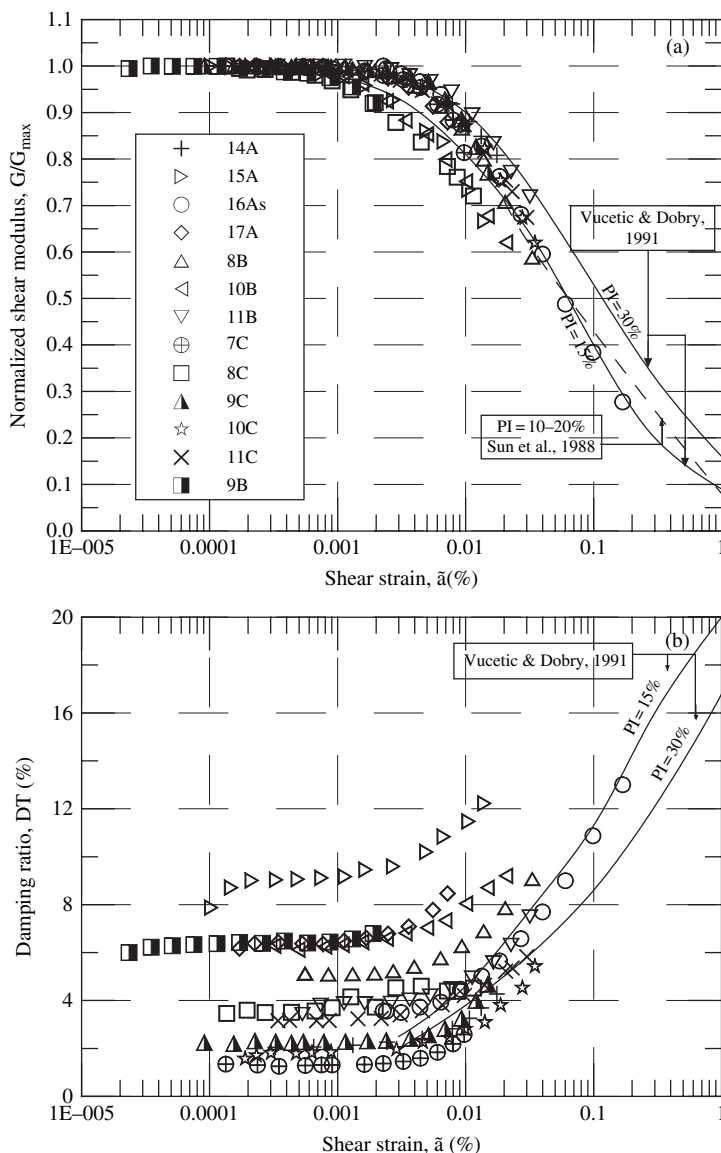


FIGURE 15 Variation of (a) normalized shear modulus, $G/G_{\max}$, and (b) damping ratio, DT , with shear strain, γ , at σ_o^{insitu} for undisturbed soils with plasticity index, $PI = 16-25\%$.

$$\frac{G}{G_{\max}} = a + b \left(0.5 + \frac{\text{atan}\left(\frac{PI-c}{d}\right)}{\pi} \right) + e \left(0.5 + \frac{\text{atan}\left(\frac{\gamma-f}{g}\right)}{\pi} \right) + h \left(0.5 + \frac{\text{atan}\left(\frac{PI-c}{d}\right)}{\pi} \right) \left(0.5 + \frac{\text{atan}\left(\frac{\gamma-f}{g}\right)}{\pi} \right) \quad (R^2 = 0.902) \quad (10)$$

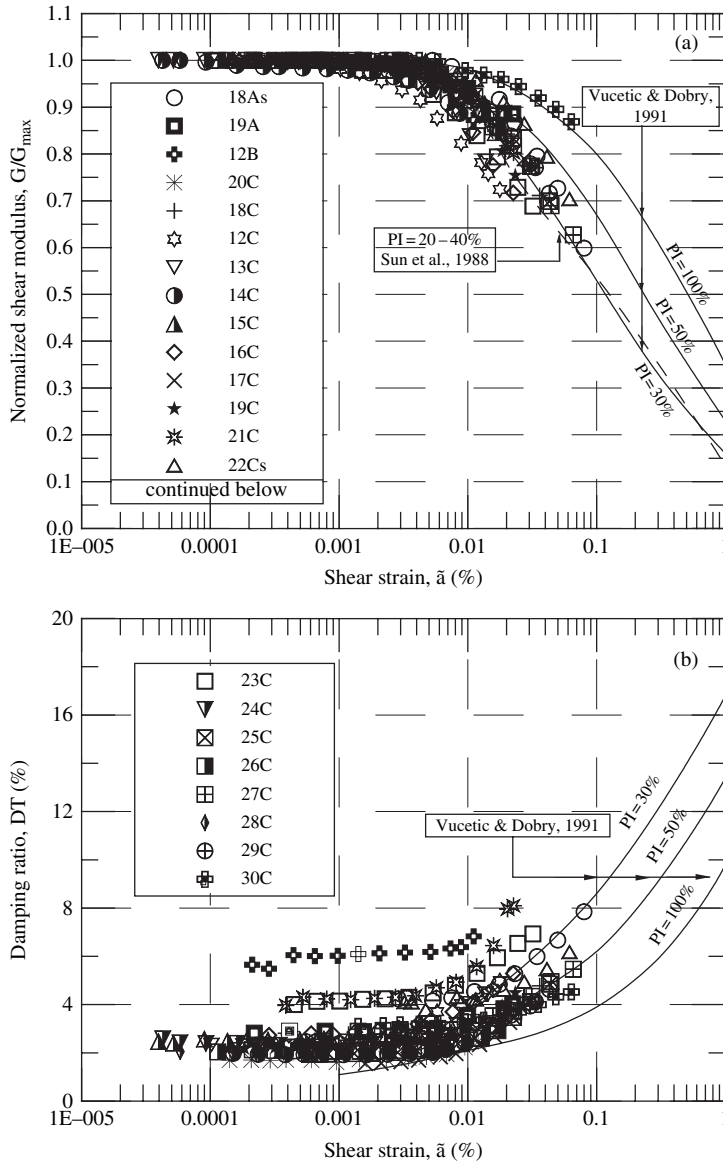


FIGURE 16 Variation of (a) normalized shear modulus, $G/G_{\max}$, and (b) damping ratio, DT , with shear strain, γ , at σ_o^{insitu} for undisturbed soils with plasticity index, $PI = 26-66\%$.

where

$$\begin{aligned} a &= 0.99418785, \quad b = -2.1598671, \quad c = 10.039495, \quad d = -16.863967, \\ e &= 0.062926143, \quad f = -0.013688113, \quad g = -0.02900694, \quad h = 5.9454009, \\ &\text{and } PI \text{ and } \gamma \text{ are expressed as percentage values (\%).} \end{aligned}$$

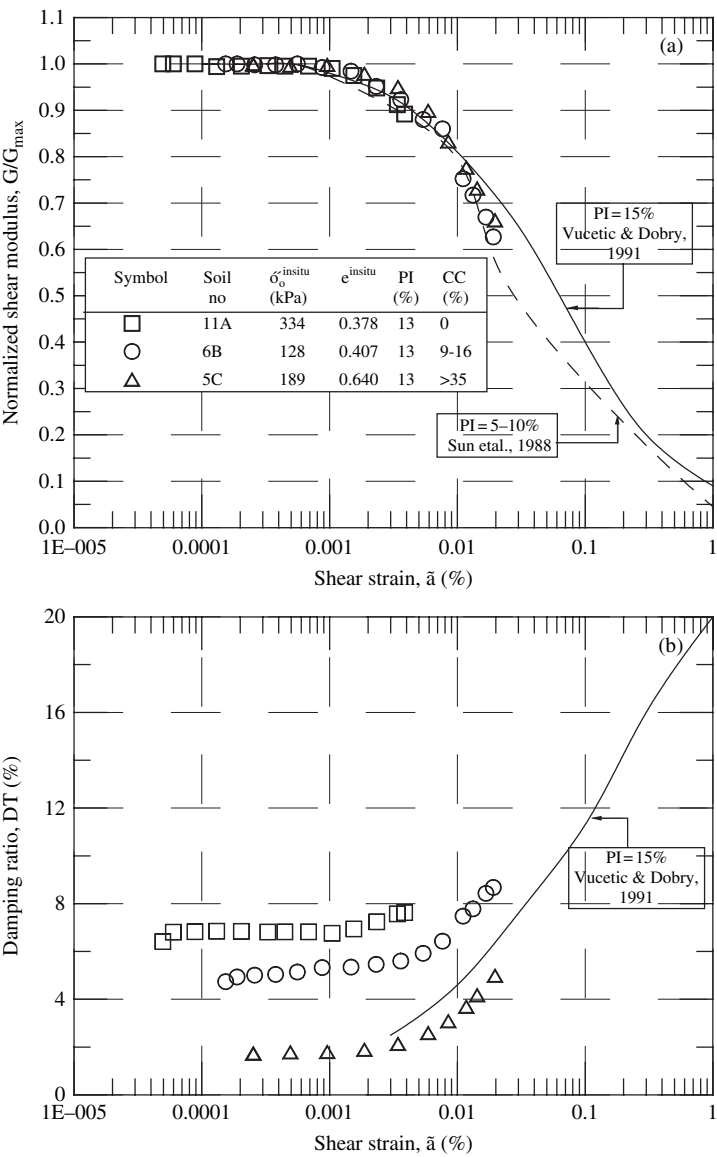


FIGURE 17 Effect of calcium carbonate content on variation of (a) normalized shear modulus, G/G_{max} , and (b) damping ratio, DT , with shear strain, γ , for three undisturbed soils.

The above equation is valid for soils with $5\% \leq PI \leq 66\%$ and $OCR = 1-16$ and values of $G/G_{max} \leq 1$.

From Eq. (10), a family curves showing the influence of PI on G/G_{max} were derived; see Fig. 19. The curves are plotted up to the maximum γ value for which test results are available. This maximum γ value changes for soils of different plasticity and varies from 2×10^{-4} to 7×10^{-4} , as indicated in the above figure. In the same figure, these curves are compared with those proposed previously [Vucetic and Dobry, 1991]. It is shown that for soils of low plasticity ($PI < 20\%$) the shear modulus degradation curves proposed in this

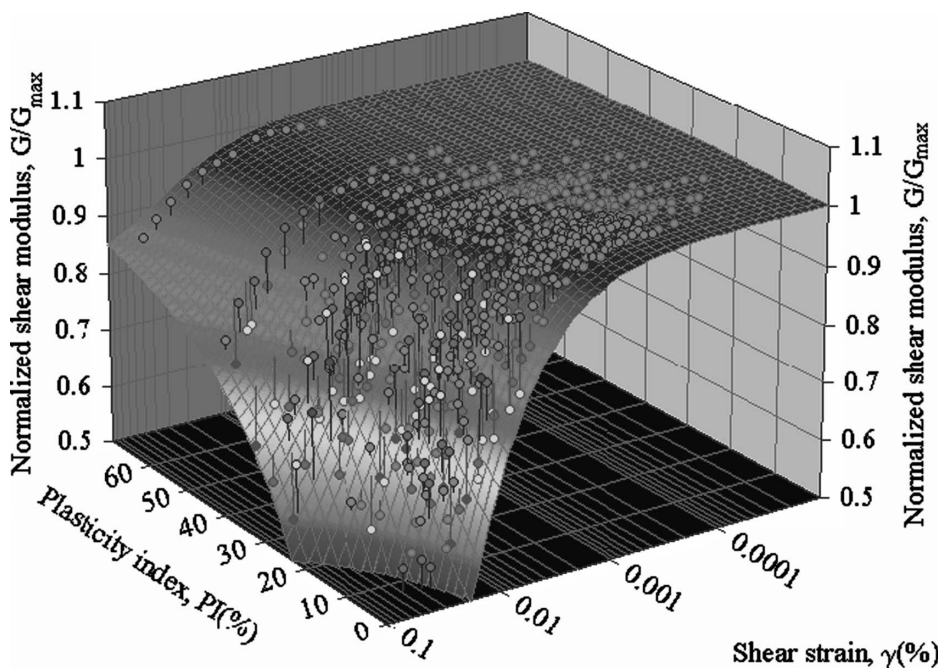


FIGURE 18 Variation of normalized shear modulus, $G/G_{\max}$, with shear strain, γ , at various PI σ_o^{insitu} for undisturbed soils and $\sigma_o' \approx 100 \text{ kPa}$ for reconstituted soils).

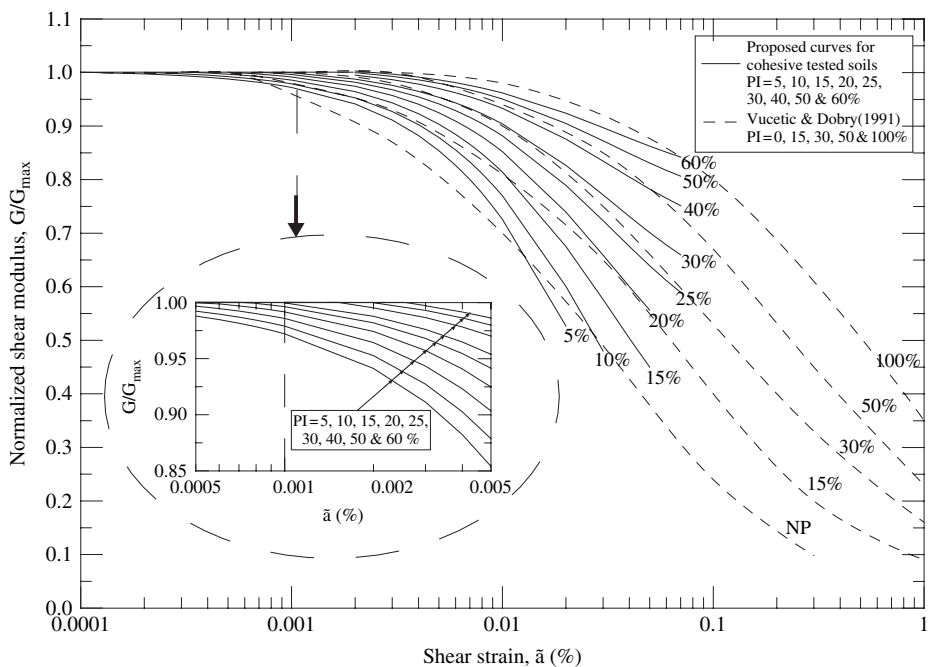


FIGURE 19 Proposed degradation curves of normalized shear modulus, $G/G_{\max}$, with shear strain, γ , at different values of PI .

article are initially flatter and then steeper than those presented by Vucetic and Dobry [1991], whereas for soils of higher plasticity, they tend to be flatter than the curves proposed by the above authors. From the family of the proposed $G/G_{\max}$ - γ curves, the variation of γ versus PI at the threshold elastic limit (considered arbitrary to correspond to $G/G_{\max} = 0.98$), as well as at lower values of $G/G_{\max}$, is derived and plotted in Fig. 20. It is shown that, as PI increases, the level of γ required to induce non-linear behavior and stiffness degradation also increases.

The relationship between the increase of DT over $DT_{\min}$ and $G/G_{\max}$, at σ_o^{insitu} , for both undisturbed and reconstituted soils, is given in Fig. 21. The data define a narrow

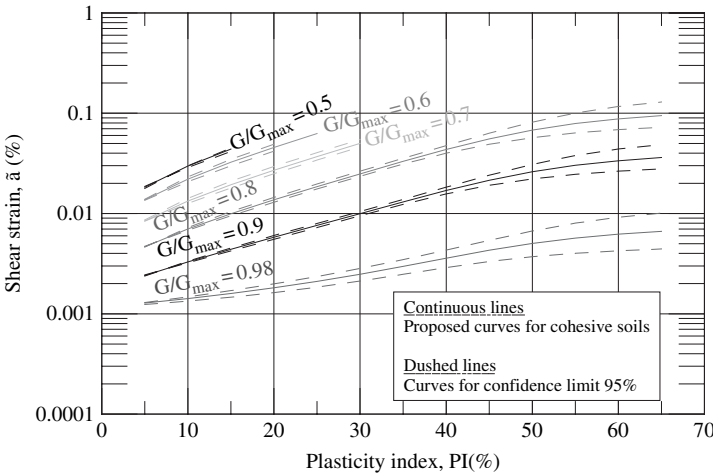


FIGURE 20 Variation of shear strain for various levels of $G/G_{\max}$ with plasticity index, PI.

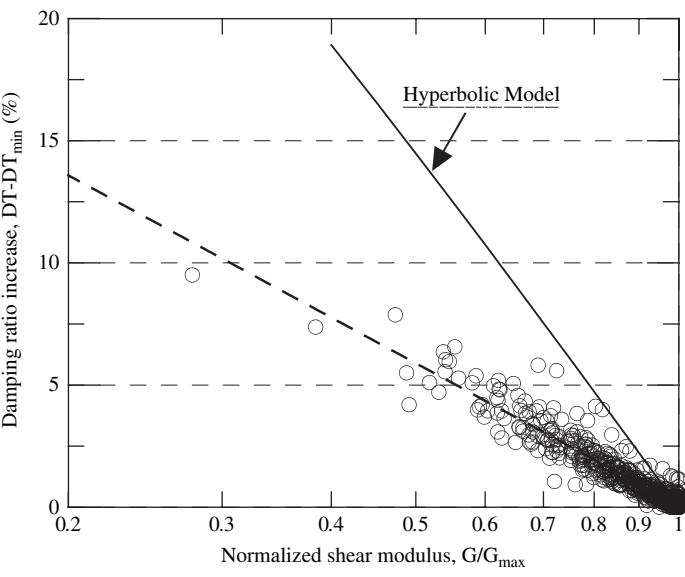


FIGURE 21 Variation of damping ratio increase, $DT-DT_{\min}$, with normalized shear modulus, $G/G_{\max}$, σ_o^{insitu} for undisturbed soils and $\sigma_o' \approx 100\text{kPa}$ for reconstituted soils).

range for which the following equation is proposed:

$$DT - DT_{\min} = 0.0741 - 8.3929 \ln \left(\frac{G}{G_{\max}} \right) \quad (R^2 = 0.91) \quad (11)$$

where

DT and $DT_{\min}$ are expressed as percentag values (%).

4.4. Practical Implications

The test results presented here allow the formulation of a soil model for the evaluation of G and DT in engineering evaluations. Since $G_{\max}$ is independent of strain rate effect, a first estimation of $G_{\max}$ for natural soils with $CC < 5\%$ can be obtained from Eq. (8) both for dynamic and static loading. Allowance should be made for the difference between laboratory and field value of $G_{\max}$; see Fig. 9. For a given soil, for which PI and e^{insitu} can be determined from the classification tests and the stress history of the site during ground investigation respectively, $DT_{\min}$ can be estimated from e^{insitu} (Fig. 12), and the $G/G_{\max}$ versus γ curve from PI (Fig. 19). For the same soil at a given γ , $DT - DT_{\min}$ can be determined as a function of $G/G_{\max}$ from Eq. (11) and then the DT versus γ variation can be obtained.

Table 8 presents the $G/G_{\max}$ and DT values, suggested by the Eurocode for the design of structures for earthquake resistance (EC8) at three levels of seismic acceleration, irrespectively of soil type and strain level. According to the test results presented here, both $G/G_{\max}$ and DT parameters are affected by the type of soil and the magnitude of shear strain. In the above table, the DT values, obtained in this work, are also included. It is shown that higher values of $DT_{\min}$, defining a broader range, are generally anticipated by the test results presented here.

Current practice in seismic site response analysis is heavily based on the $G/G_{\max}$ and DT versus γ curves, published by Vucetic and Dobry [1991]. The quantification of the observed differences between the $G/G_{\max}$ versus γ curves proposed in this work and those presented by Vucetic and Dobry [1991] requires a detailed parametric site response analysis. As far as the damping ratio is concerned, it may be anticipated qualitatively

TABLE 8 Comparison of $G/G_{\max}$ and DT values proposed by EC 8 and corresponding values determined by RC tests

EUROCODE 8			RC tests	
Design acceleration (g)	$G/G_{\max}$	DT (%)	$G/G_{\max}$	DT (%)
0.1	0.8 (± 0.1)	3	0.9	1.3–12.3
			0.8	1.8–12.3
			0.7	2.2–12.3
0.2	0.5 (± 0.2)	6	0.6	2.6–12.3
			0.5	4.8–>12.3
0.3	0.3 (± 0.2)	10		

that high values of $DT_{\min}$, as observed for some soils with low void ratios ($e < 0.5$), may result in a reduction of dynamic amplification of such soil deposits, when the behavior remains within the linear range during earthquake shaking.

5. Conclusions

The main conclusions of the work presented are as follows:

1. The small-strain shear modulus, $G_{\max}$, is mainly influenced by plasticity, void ratio, stress history, mean effective stress, and structure (fabric and bonding) of soil. Equations (7) and (8) are proposed for the estimation of $G_{\max}$ of reconstituted and undisturbed natural soils with $CC < 5\%$, respectively. Normalized shear modulus, $\frac{G_{\max}}{f(e)g(PI)}$, for undisturbed natural soils with $CC < 5\%$ is higher than the corresponding for reconstituted soils due to effect of ageing and fabric. Natural soils with $CC \geq 5\%$ have either similar or greater values of $\frac{G_{\max}}{f(e)g(PI)}$ than soils with $CC < 5\%$. Similar values of $\frac{G_{\max}}{f(e)g(PI)}$ indicate that the stiffness component due to cementation is either insignificant or lost during the sampling process, whereas higher values of $\frac{G_{\max}}{f(e)g(PI)}$ imply that strong interparticle bonding might have been developed.
2. The most important parameter controlling $DT_{\min}$ is soil structure. For soils having void ratios greater than 0.5 at σ_o^{insitu} , $DT_{\min}$ is of the order of 1.4–4.7%. However, for soils with a void ratio less than 0.5 at σ_o^{insitu} , $DT_{\min}$ varies from 2.2–8.7%. The high values of $DT_{\min}$ for some of the soils with $e < 0.5$ may be attributed to the sensitivity of their structure to strain-rate, due to increased interaction of soil particles resulting in high dissipation of energy at particle contacts.
3. Plasticity index is the main parameter, controlling the degradation of shear modulus at high strains. Equations (10) and (11) are proposed to express the influence of PI on the variation of $G/G_{\max}$ with γ , as well as the increase of damping ratio over its small-strain value, $DT-DT_{\min}$, with $G/G_{\max}$, respectively. Comparison between the $G/G_{\max}$ - γ curves proposed in this work with the corresponding proposed by Vucetic and Dobry [1991], shows that for soils of low plasticity ($PI < 20\%$) the degradation curves proposed in this paper are initially flatter and then steeper, whereas for soils of higher plasticity, they tend to be flatter than those proposed by the above authors.
4. The test results allow the formulation of a soils model for the evaluation of G and DT both at small and high strains for practical use in seismic site response analysis.

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